

Cool by day, warm by night: surface heat, yearly land cover and people in metropolitan Kano, 2015 to 2026

Eight local government areas, Landsat, MODIS and GRID3

Wisdom Akpabio · October 2026

Surface temperature and land cover, adapted from [Ramachandra, Rana, Vinay and Aithal, *Scientific Reports* \(2025\)](#).

Code and data: <https://github.com/whizialfa/kano-surface-heat>

ABSTRACT

Kano is cooler than the farmland around it by day and warmer by night. In all 28 clear Landsat scenes from March and April between 2015 and 2026, the metropolitan surface at about 10:30 was cooler than a ring 1 to 10 km outside it (median $-1.6\text{ }^{\circ}\text{C}$), and the new fringe of Ungogo and Kumbotso was hotter than the old city (median $+1.75\text{ }^{\circ}\text{C}$). MODIS passes at 22:30 and 01:30 flip it: in the hot season the city is 1.4 and $1.1\text{ }^{\circ}\text{C}$ warmer than its ring, and the old city becomes the warmest ground. Built-up cover is Kano's cool daytime ground: each extra built-up 10 m cell in a 30 m pixel lowers the surface, from $47.0\text{ }^{\circ}\text{C}$ with none to $44.8\text{ }^{\circ}\text{C}$ with all nine, while farmland and bare earth raise it and greenness (NDVI) predicts nothing. This reverses the Bangalore result that the analysis adapts (Ramachandra et al., 2025). Most people live on the cooler ground, but of the 276,000 people on the hottest fifth of populated ground, **95%** are also more than 15 minutes' walk from five clinics and five schools. Built-up land grew from 38% to 52% of the metropolitan area, and farmland that became city cooled by day by 0.3 to $0.6\text{ }^{\circ}\text{C}$ relative to farmland that stayed open, on independent land cover. These are surface temperatures, not air temperatures.

KEYWORDS

land surface temperature; surface cool island; night-time heat; land cover; Landsat; MODIS; Kano; Sahel

Glossary

Land surface temperature (LST)

How hot the ground, roof or canopy is, measured from orbit. Not the air temperature people feel.

Rural ring

Land 1 to 10 km outside the study outline, used as the reference for the city.

Surface cool island

A city whose surface is cooler than its surroundings. Kano is one by day in the dry seasons and a heat island by night.

Old city and fringe

Old city: Kano Municipal, Dala, Gwale, Fagge, Nassarawa and Tarauni. Fringe: Ungogo and Kumbotso.

Composition class

The anchor paper's twelve labels (C1 to C12) for the mix of built-up, bare, green and water 10 m cells inside one 30 m pixel.

NDVI, NDBI, MNDWI, bare-soil index

Spectral indices for greenness, built-up surface, water and moisture, and bare soil.

MODIS composite

An 8-day average of clear-sky surface temperature at 1 km, from the Terra (about 10:30 and 22:30) or Aqua (about 13:30 and 01:30) satellite.

UTFVI

Urban thermal field variance index: each pixel's LST relative to the city mean. On the standard scale in kelvin, 1.7% of Kano is "strong" or worse (Bangalore reports 76% unfavourable). The index is relative to each city's own mean, so shares do not compare across cities.

Walk time (PT_k)

Walking time, in minutes at 5 km/h, to the five nearest clinics and the five nearest schools, averaged. From *Fifteen minutes on foot*.

1. Introduction

Cities are usually written as heat islands: the built surface hotter than the country around it, day and night (Oke, 1982; Voogt and Oke, 2003). That is the ordinary story where trees and wet soil are replaced by roofs (Peng et al., 2012). In dry cities the sign can flip. Bare fields heat faster than compounds and courtyards, so the city is the cooler ground at midday and the warmer ground after dark (Lazzarini, Marpu and Ghedira, 2013; Rasul, Balzter and Smith, 2015). Kano sits in that second group for most of the year. A good deal of the advice that still reaches Nigerian planners is written as if it sat in the first.

Ramachandra, Rana, Vinay and Aithal (2025) take one April morning over Bangalore, classify the mix of built-up, bare, green and water inside each 30 m Landsat pixel, and recommend at least 30% vegetation per plot. Bangalore in April is humid and green. Kano in March is not. A single overpass also cannot say whether the pattern holds after sunset, or whether twelve years of building on the fringe has cooled the daytime surface or heated it.

This paper asks the same composition question of metropolitan Kano and adds what that morning scene left out: every clear hot-season Landsat picture from 2015 to 2026, night-time MODIS passes, a land-cover map for each year, the people on the hexes, and the walking times already measured in *Fifteen minutes on foot* (Akpabio, 2026). It is an adaptation of Ramachandra et al. The departures are listed in the repository's metric trace.

2. Statement of the problem

The sentence that travels with "urban heat" is that cities are hotter than their countryside, and that more vegetation will cool them. Run that sentence on Kano in the hot season and it fails twice. The metropolitan surface at 10:30 is cooler than the farmland 1 to 10 km outside it, in every clear March and April scene on file. Greenness, as a satellite measures it, does not pick the cool ground. Built-up cover does: walls, roofs, courtyards and the shade they throw.

A second problem sits on the same map. The people on the hottest daytime ground are not in the old city. They are on the farmland edge in Ungogo and Kumbotso, and they are also the people more than fifteen minutes' walk from clinics and schools. Advice written for Bangalore, or for a city treated as one heat island, will spend the shade in the wrong place and at the wrong time of day.

The question this paper can actually answer is narrower. On a clear morning in the hot season, is metropolitan Kano hotter or cooler than its ring? What covers the hot pixels? Who lives on them, and how far do they walk? After dark, does the ranking of old city and fringe reverse? And when farmland is built over, does the daytime surface cool or heat relative to farmland that stays open?

3. Study area

The city used here is the same eight local government areas as *Fifteen minutes on foot*: Dala, Fagge, Gwale, Kano Municipal, Kumbotso, Nassarawa, Tarauni and Ungogo. Together they are 573 km² and 5.78 million people on the GRID3 / WorldPop Nigeria v3.0 grid, scaled to July 2025. A score that stopped at the walled city and Sabon Gari would describe a Kano that no longer exists. Ungogo to the north and Kumbotso to the south now hold the later streets and a large share of the people.

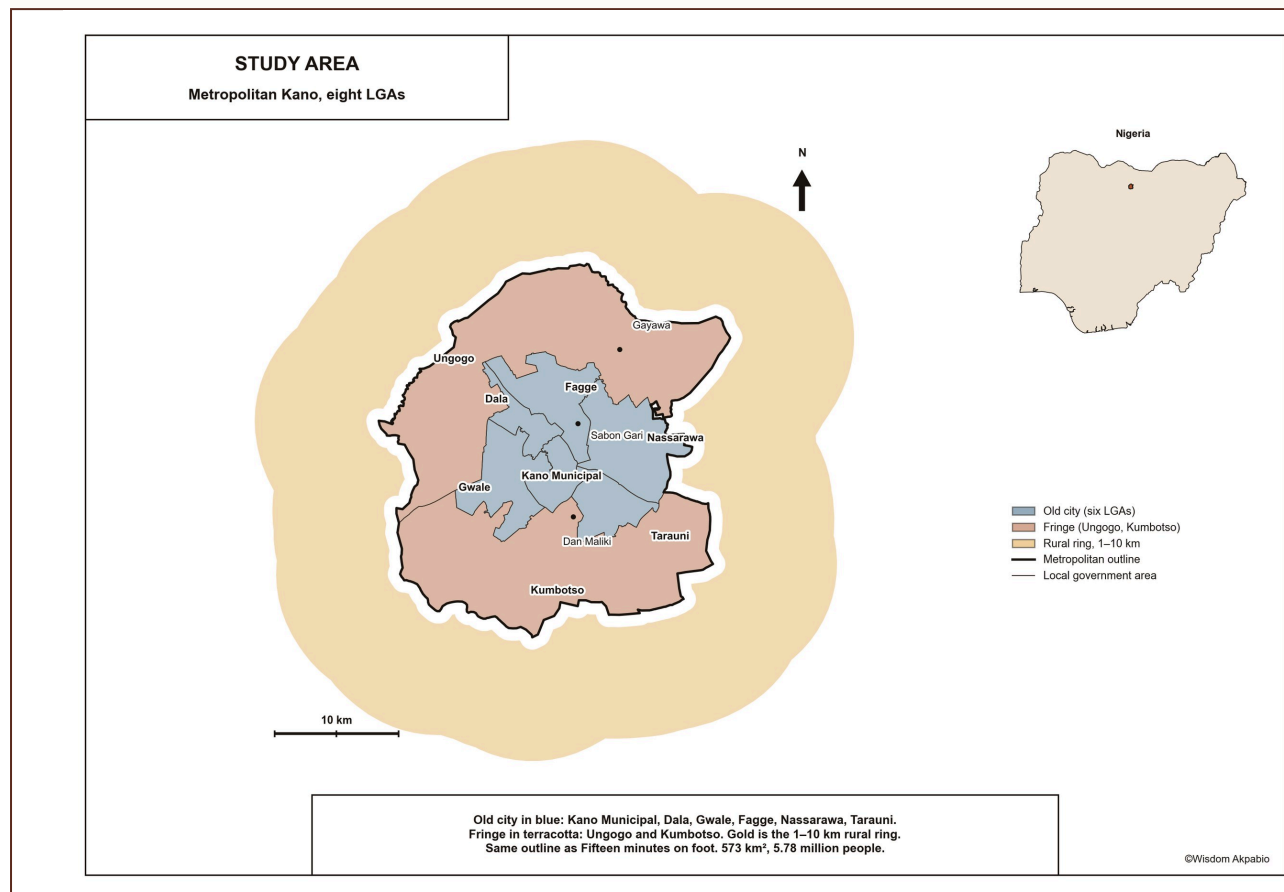


Figure 1. Metropolitan Kano as used in this paper. Blue is the old city (six local government areas). Terracotta is the fringe, Ungogo and Kumbotso. Gold is the rural ring, 1 to 10 km outside the outline. 573 km², 5.78 million people. Same outline as *Fifteen minutes on foot*.

Those eight areas are split for the heat work. The old city is Kano Municipal, Dala, Gwale, Fagge, Nassarawa and Tarauni. The fringe is Ungogo and Kumbotso. The rural reference is a ring 1 to 10 km outside the metropolitan outline: far enough that the comparison is farmland, close enough that it is

the same weather. Figure 1 is that cut. The outline, wards, 200 m hexagons and named settlements are shared with the walking paper. Areas are measured on the study-boundary file in UTM zone 32N.

4. Data and methods

4.1 Surface temperature

Landsat 8 and 9 Collection 2 Level-2 surface temperature (USGS single-channel algorithm, atmospherically corrected, ASTER emissivity), from Microsoft Planetary Computer. Of 1,160 scenes searched, 142 cover at least 98% of the outline with at least 98% of those pixels clear in the QA band. One footprint (WRS-2 path 188, row 52) covers the whole city. The hot season is March to May: 40 scenes, one to six a year. Each year's surface is the per-pixel median of its scenes. The pooled surface is the median of the twelve yearly medians, so a year with six clear days does not outvote a year with one. Figure 4 maps those yearly surfaces as a contrast to that year's ring, so a hot or cool year does not recode the city.

4.2 Land-use mix

ESA WorldCover 10 m (2021), warped onto a 10 m grid that nests exactly three by three inside the 30 m LST grid, and collapsed to the anchor's four groups: built-up, bare, vegetation, water. Dry-season cropland is bare earth to a thermal sensor, so the analysis is run twice: cropland as vegetation (the anchor's rule) and cropland as bare. The anchor's twelve composition classes overlap and leave gaps; they are assigned in a fixed order and the unassigned share is reported. Because that table is ambiguous, the main land-use result is a gradient that needs no classes: how temperature moves as the nine 10 m cells inside a 30 m pixel fill with built-up, farmland or greenery.

4.3 Spectral indices

NDVI, NDBI, MNDWI and the bare-soil index from the same scenes, medians per season. As in the anchor: 1,000 random pixels, simple correlations and a four-index linear model. Added here: the same model on all 636,483 pixels, variance inflation factors, and Moran's I of the residuals.

4.4 Land cover for every year

WorldCover is a single 2021 map. For each year from 2015 to 2026 the land cover is classified from the same clear Landsat scenes: hot-season and wet-season reflectance composites (six bands each) with NDVI, NDBI, MNDWI and the bare-soil index, because dry-season farmland is bare in March and green in August. A random forest is trained on 30 m cells whose nine WorldCover cells agree in both 2020 and 2021 (six classes: built-up, cropland, bare ground, grass and shrub, trees, water). Spatially blocked 3 km cross-validation gives 85.6% overall accuracy (kappa 0.83); built-up F1 0.87, grass and shrub weakest at 0.71. Class probabilities are averaged over three years, which fills cloud gaps and damps one-year flips. Impact Observatory's annual 10 m maps (Sentinel-2, 2017 to 2023) are the independent check. Figure 2 is the accuracy. The yearly maps come later, with the change result.

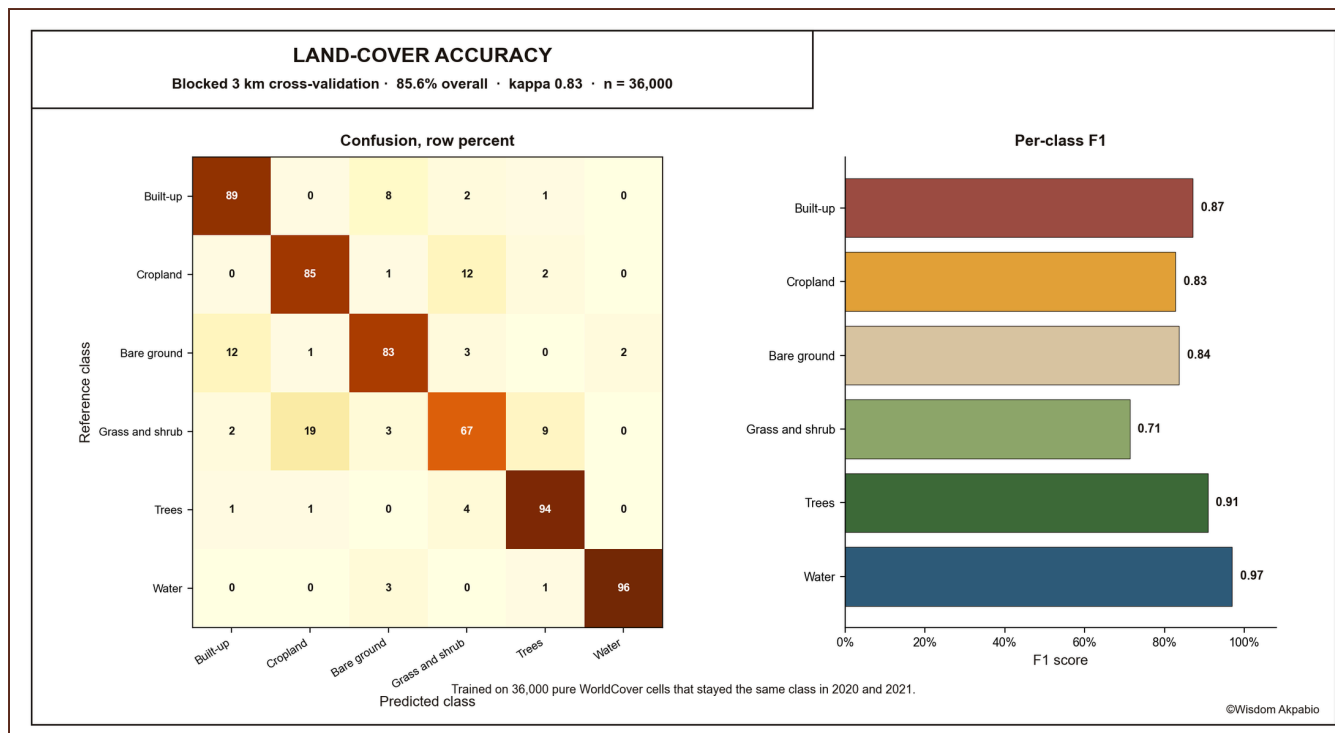


Figure 2. Accuracy of the yearly classification. Left: blocked cross-validation confusion, row percent. Right: per-class F1. Overall accuracy 85.6%, kappa 0.83, on 36,000 pure WorldCover cells that stayed the same class in 2020 and 2021. Grass and shrub is the weak class.

4.5 Day and night

Landsat sees Kano only at about 10:30. MODIS MOD11A2 (Terra, about 10:30 and 22:30) and MYD11A2 (Aqua, about 13:30 and 01:30) 8-day clear-sky composites at 1 km, 2015 to 2026, all seasons, give four times of day. They are warped onto a 1 km version of the same grid, outline, ring and old-city/fringe split. A composite counts only when at least 80% of both city and ring pixels are valid. 1,022 composites.

4.6 People and walking

Hexagon population from GRID3 v3.0; under-5, over-65 and women 15 to 49 from the v3.0 age-sex layers. Each hexagon carries its walking time to five clinics and five schools from the walking paper. The clock there is 5 km/h on mapped streets. It does not know if there is a walkway, if the street floods, or if anyone would take it in the afternoon heat.

5. Results

5.1 A city cooler than its fields

Table 1. City against its rural ring, and fringe against old city, in every clear hot-season Landsat scene, 2015 to 2026. Differences in surface temperature at about 10:30.

Scenes		City cooler than ring	Median city minus ring	Fringe hotter than old city	Median fringe minus old city
March and April	28	28	−1.6 °C	28	+1.75 °C
May	12	9	−1.0 °C	10	+1.3 °C
All hot season	40	37	−1.5 °C	38	+1.6 °C

The three scenes where the city is warmer than its ring are 1 May 2018 (+0.05 °C), 4 May 2019 (**+4.3 °C**) and 28 May 2025 (+1.7 °C). On 4 May 2019 the ring fell to 42.4 °C while the city held 46.7 °C, the signature of wet fields after an early storm. That one scene carries the only positive yearly median in the series (Figure 3).

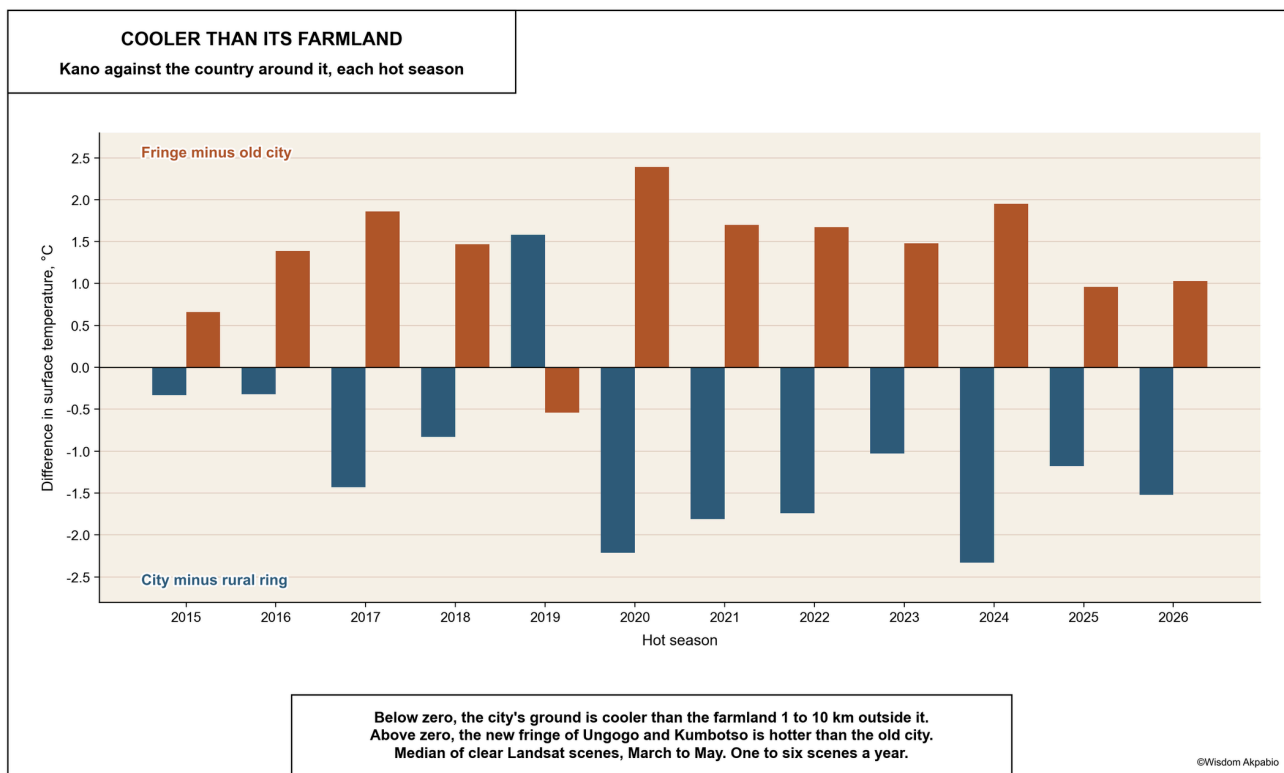


Figure 3. Each hot season, the city minus its rural ring (below zero: the city is cooler) and the fringe minus the old city (above zero: the fringe is hotter). 2019 is the one year an early-May storm tips the median.

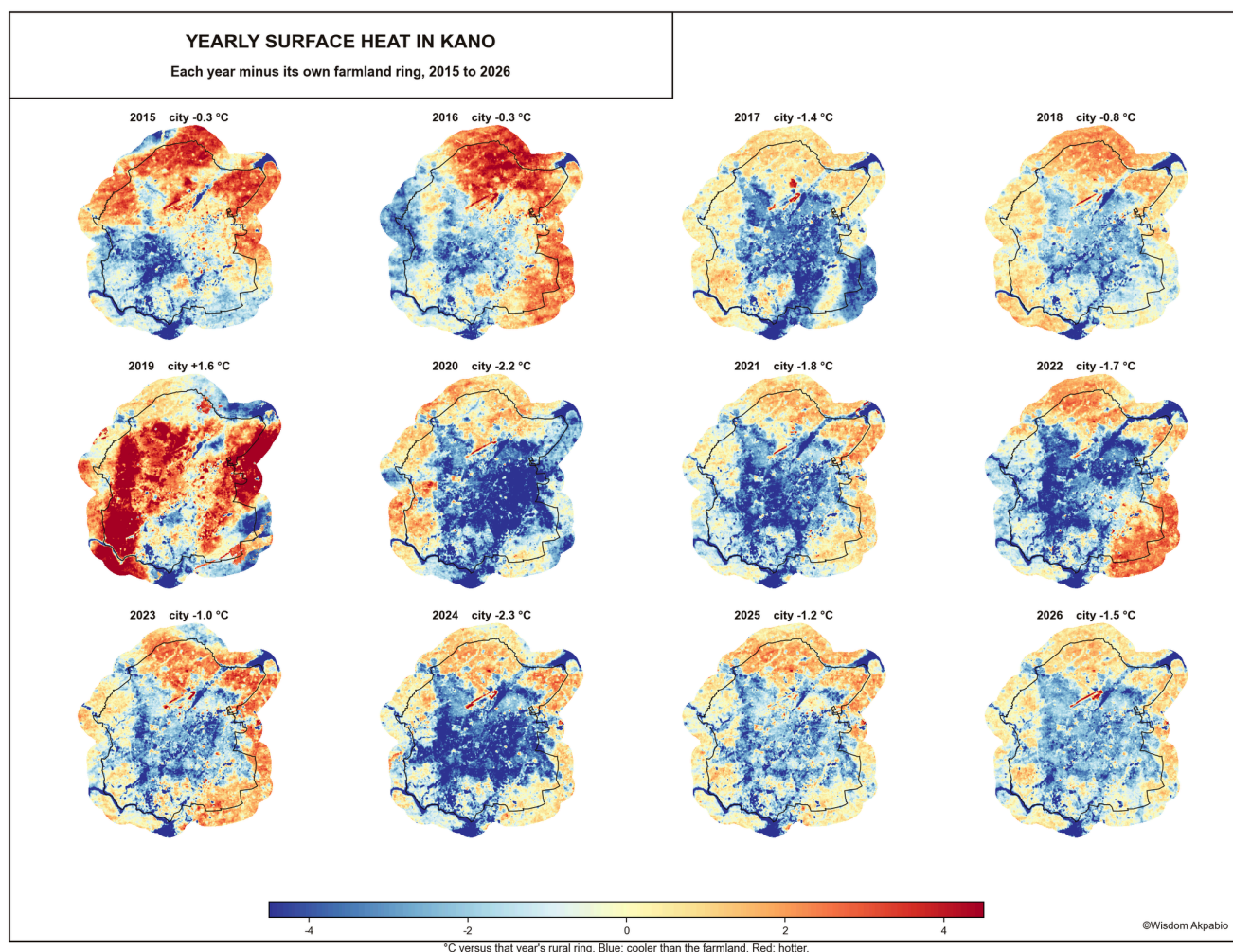


Figure 4. Hot-season surface each year, minus that year’s rural-ring median. Blue is cooler than the farmland around the city; red is hotter. A shared scale so a hot year does not recode the map. 2019 is the wet-field year.

Across the pooled surface, local government medians run from Kano Municipal (**44.0 °C**) and Gwale (44.2) to Fagge (45.8) and Ungogo (**46.9 °C**). Hot spots (above the city mean plus two standard deviations, 49.8 °C) cover **1.7 km²**, and **1,854** of their 1,922 pixels are cropland or grass, not buildings: Fanisau, Rangaza and Karo in Ungogo, and Kwachiri in Fagge, the ward that contains Mallam Aminu Kano International Airport and its runway. Cold spots (below 42.2 °C) cover **14.3 km²**, along the Challawa river and the dense southern wards of Panshekara, Naibawa, Dorayi and Kumbotso (Plate 1).

5.2 What covers the hot ground

Table 2. Mean hot-season surface temperature by the anchor’s composition classes, pooled 2015 to 2026. Share of the city’s 30 m pixels in brackets.

Composition class	Bangalore	Kano, cropland as vegetation	Kano, cropland as bare
C1, all bare	39.92 °C	44.67 °C (0.7%)	47.39 °C (34.5%)
C2, all built-up	39.74 °C	44.75 °C (38.0%)	44.75 °C (38.0%)
C11, all vegetation	cool	47.11 °C (47.2%)	45.85 °C (6.5%)
C12, all water	coolest	39.08 °C (0.05%)	39.08 °C (0.05%)
Unassigned	not reported	4.0%	5.5%

Read the anchor’s way, Kano’s all-vegetation pixels are its hottest ground, because most of what WorldCover calls vegetation is cropland that is bare in March. Read the Sahel’s way, bare ground is hottest, as in Bangalore, but pure built-up pixels are still **1.1 °C** cooler than the remaining trees, shrub and grass, and **2.6 °C** cooler than bare fields. Only open water is cooler than the built city.

Built-up covers **45%** of the metropolitan area in WorldCover, cropland **40%**, grass 10% and trees 2% (Plate 2). Set beside Plate 1, the dull-red built-up core is the cool blue and the gold cropland ring is the hot red. The gradient in Figure 5 shows the same thing without classes.

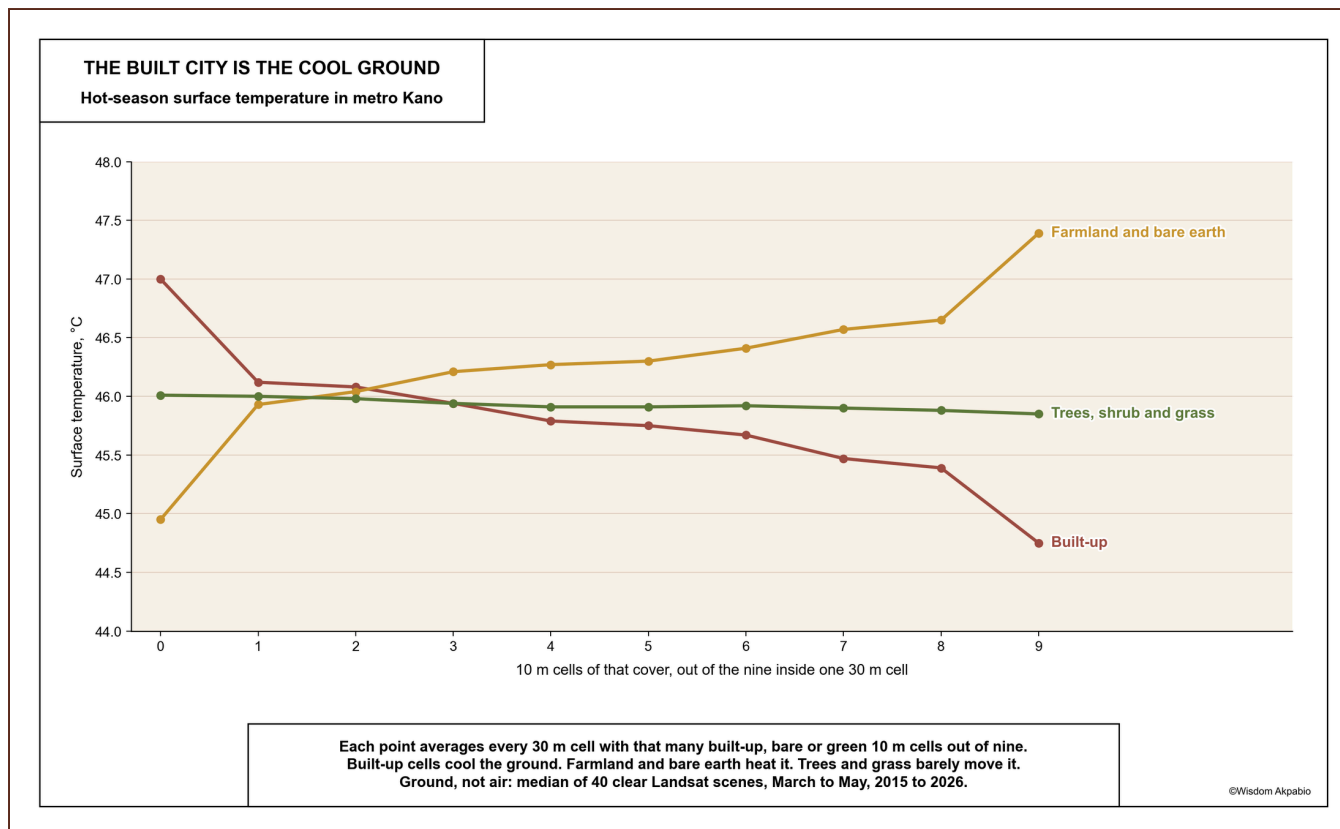


Figure 5. Hot-season surface temperature by how many of the nine 10 m cells inside a 30 m pixel are built-up, farmland or bare earth, or trees, shrub and grass. Built-up cells cool the ground; farmland heats it; greenery barely moves it.

5.3 Greenness does not predict heat here

Table 3. Correlation of hot-season surface temperature with four spectral indices, all pixels in the outline.

	NDVI	NDBI	MNDWI	Bare-soil index
Bangalore (anchor)	-0.46	+0.69	-0.23	+0.65
Kano, pooled hot season	+0.15	+0.48	-0.62	+0.53
Kano, range over 12 years	-0.04 to +0.20	+0.26 to +0.52	-0.21 to -0.64	+0.23 to +0.58

Moisture, not greenness, tracks the cool ground: the water index is the strongest single correlate in every year. The anchor's four-index equation does not transfer. The indices are collinear (variance inflation up to 34), the fitted coefficients change sign between years (the bare-soil term runs from +12 to -87), the model explains about half the variance (R^2 0.50 pooled) and its residuals are strongly clustered in space (Moran's I 0.30 to 0.67). The simple correlations are reported. The model is not.

5.4 Who lives on the hot ground

Table 4. People by hot-season surface temperature of their 200 m neighbourhood. Populated: at least 50 people in the hexagon.

	People	Under 5	Over 65
Whole city	5,780,865		
Hottest tenth of populated hexagons ($\geq 47.7^{\circ}\text{C}$)	122,404 (2.1%)	20,529	2,675
Coollest tenth ($\leq 43.6^{\circ}\text{C}$)	616,189 (10.7%)	104,972	13,676
Hottest fifth ($\geq 47.1^{\circ}\text{C}$) and more than 15 minutes' walk	260,964	42,467	
Hottest fifth and within 15 minutes' walk	14,873	2,362	

People live on the cooler ground: the population-weighted surface temperature is **45.0 °C**, a full degree below the average hexagon (46.0 °C). The heat that does reach homes sits on the farmland edge, and almost everyone there is also far from services. Of the 276,000 people on the hottest fifth of populated ground, **95%** are more than 15 minutes' walk from five clinics and five schools; only 25 of those 831 hexagons are within a 15-minute walk (Plate 3). Across the city, hotter hexagons are farther from services (population-weighted $r = 0.36$).

The GRID3 age-sex layers apply one age structure across the city: under-fives are 16.5 to 16.7% of every local government area and over-65s 2.2%. They convert people into counts of children and elders, but they cannot show whether children live on hotter ground than adults. Counts are given. An age-weighted temperature is not.

Table 5. The five hottest populated wards and two of the coolest. Surface temperature and walking time weighted by population.

Ward	LGA	People	Surface	Walk	Largest named place
Karo	Ungogo	20,919	47.9 °C	31 min	Muncika Kudu
Yada Kunya	Ungogo	51,646	47.4 °C	25 min	Layin Kasuwa
Chalawa	Kumbotso	39,604	47.1 °C	36 min	Kusuba Kwari
Rangaza	Ungogo	114,696	46.9 °C	21 min	Layin Masallacin Juma'a
Fanisau	Ungogo	48,494	46.1 °C	23 min	Kan Fako
Dan Maliki	Kumbotso	264,515	43.9 °C	11 min	Gidan Leda
Daurawa	Tarauni	17,024	43.8 °C	8 min	Layin Barwa

Kumbotso is not uniformly hot. Its dense inner wards, Dan Maliki and Naibawa, are among the coolest ground in the city. The heat is where farmland still meets new building.

5.5 Cool by day, warm by night

Table 6. City minus rural ring by season and satellite pass, median of MODIS 8-day composites, °C. In brackets, the share of composites in which the city was cooler.

Season	10:30	13:30	22:30	01:30
Hot, March to May	−0.33 (74%)	−0.38 (74%)	+1.39 (5%)	+1.14 (9%)
Harmattan, November to February	+0.04 (44%)	−0.40 (75%)	+1.90 (0%)	+1.88 (1%)
Wet, June to September	+1.49 (6%)	+1.31 (13%)	+0.10 (31%)	+0.10 (34%)

In both dry seasons the city is the cooler ground by day and the warmer ground by night. Bare fields lose their heat quickly after sunset; walls, roofs and paving give theirs back slowly. The same flip shows inside the outline: in the hot season the fringe is 0.1 to 0.3 °C hotter than the old city by day and **1.7 to 1.8 °C** cooler at night, cooler in **99%** of night composites. In the wet season the fields are green and wet, and the city is warmer by day (**1.3 to 1.5 °C**) and roughly level at night: the textbook heat island, for four months of the year.

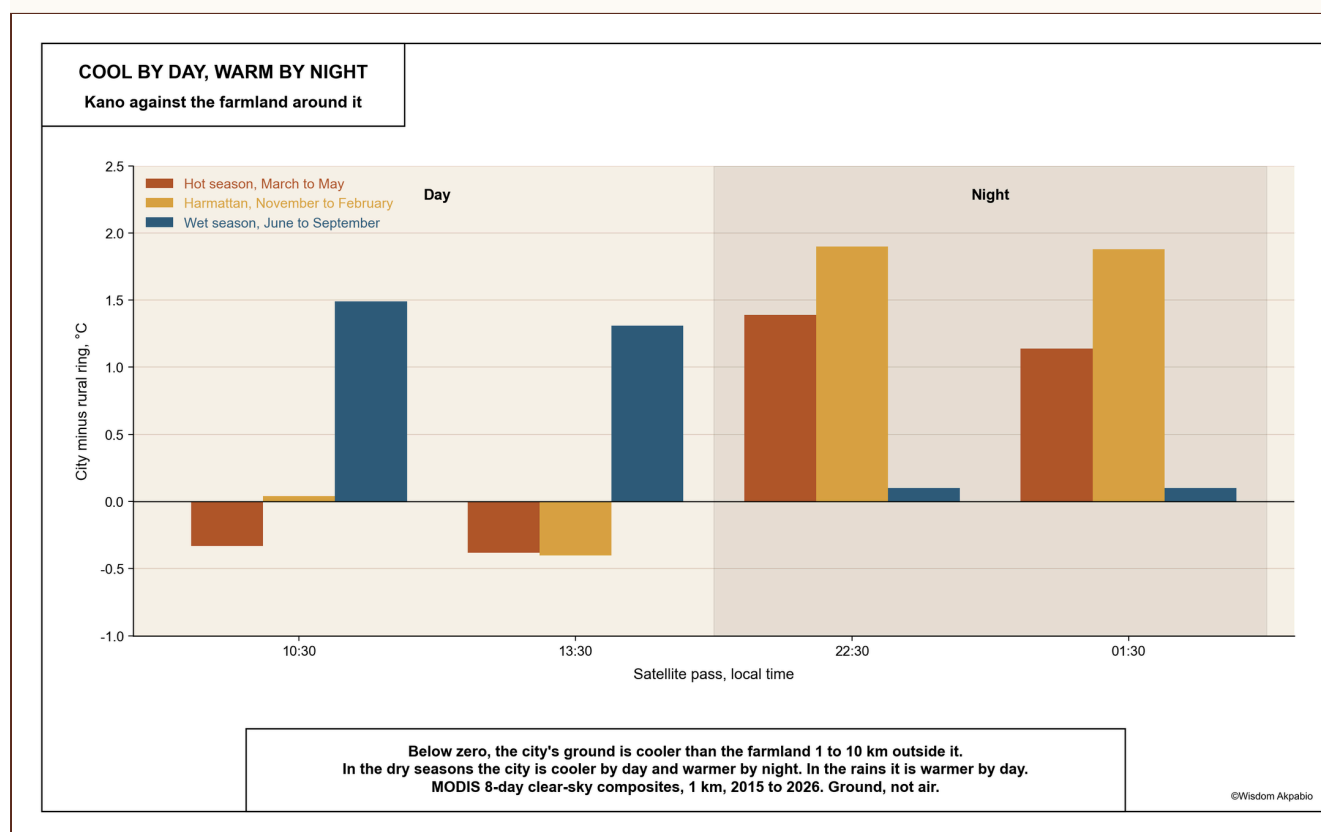


Figure 6. City minus rural ring at 10:30, 13:30, 22:30 and 01:30. In the dry seasons the city is cooler by day and warmer by night; in the rains it is warmer by day.

MODIS agrees with Landsat on the sign of the daytime contrast but gives a smaller value (**−0.3** against **−1.6 °C**). A 1 km pixel mixes compounds, fields and villages that Landsat separates at 30 m, and an 8-day composite averages several days and viewing angles. Landsat is quoted for the daytime size. MODIS is quoted for the day-night flip.

Plate 3 is a daytime map. At night the ranking of places reverses: the dense wards that are coolest at 10:30 are the warmest at 01:30, and the hot-and-far fringe cools fastest.

5.6 Building on farmland cools the ground by day

Built-up land grew from **38%** of the metropolitan area in 2015 to **52%** in 2026 on the smoothed Landsat maps (Figure 7, Figure 8). Impact Observatory puts it at 47% in 2017 and 59% in 2023: higher, but moving the same way. Growth is quoted as a range backed by both.

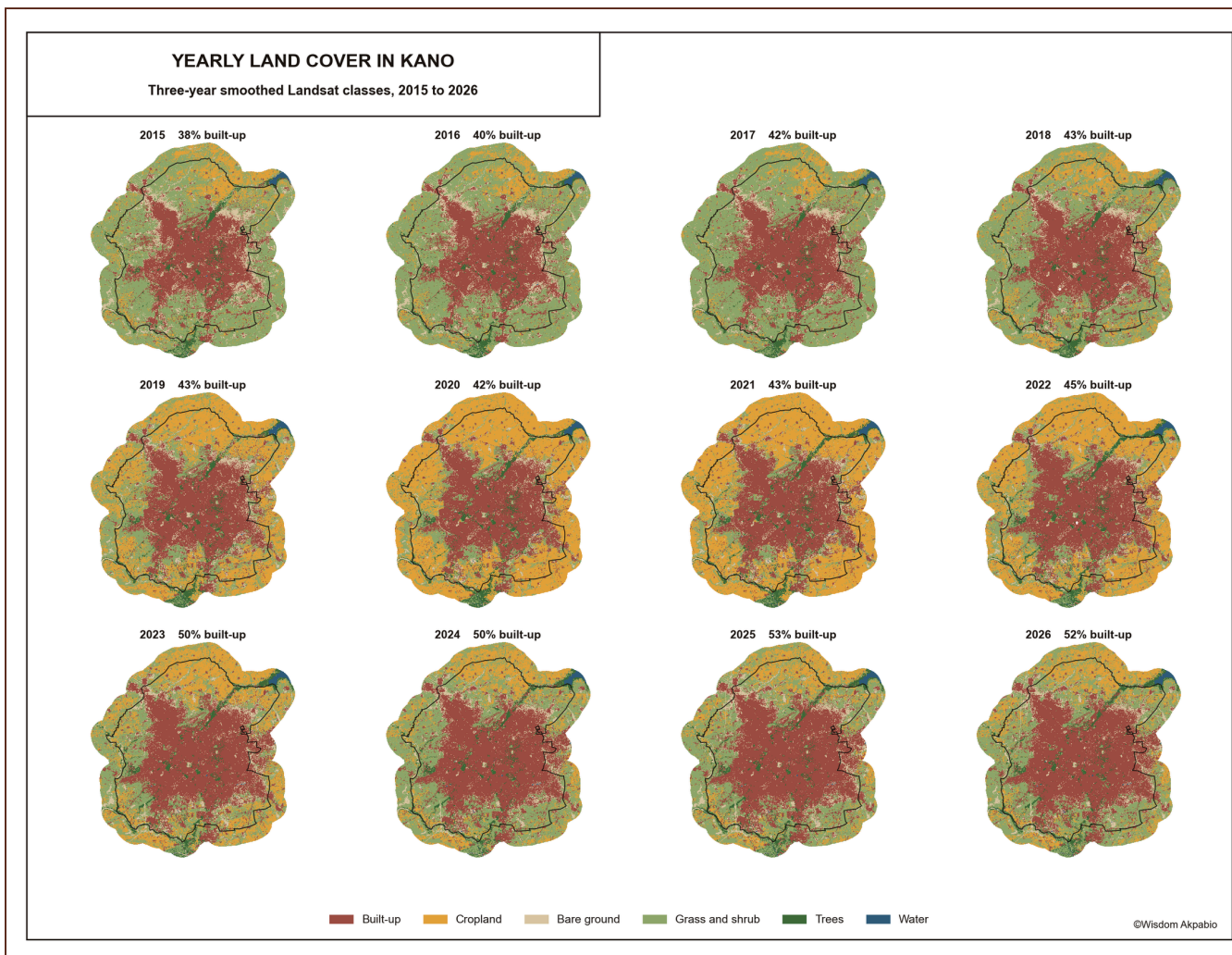


Figure 7. Yearly land cover in metropolitan Kano, 2015 to 2026. Three-year smoothed Landsat classification. Built-up is dull red. The share of the outline that is built-up is printed on each panel.

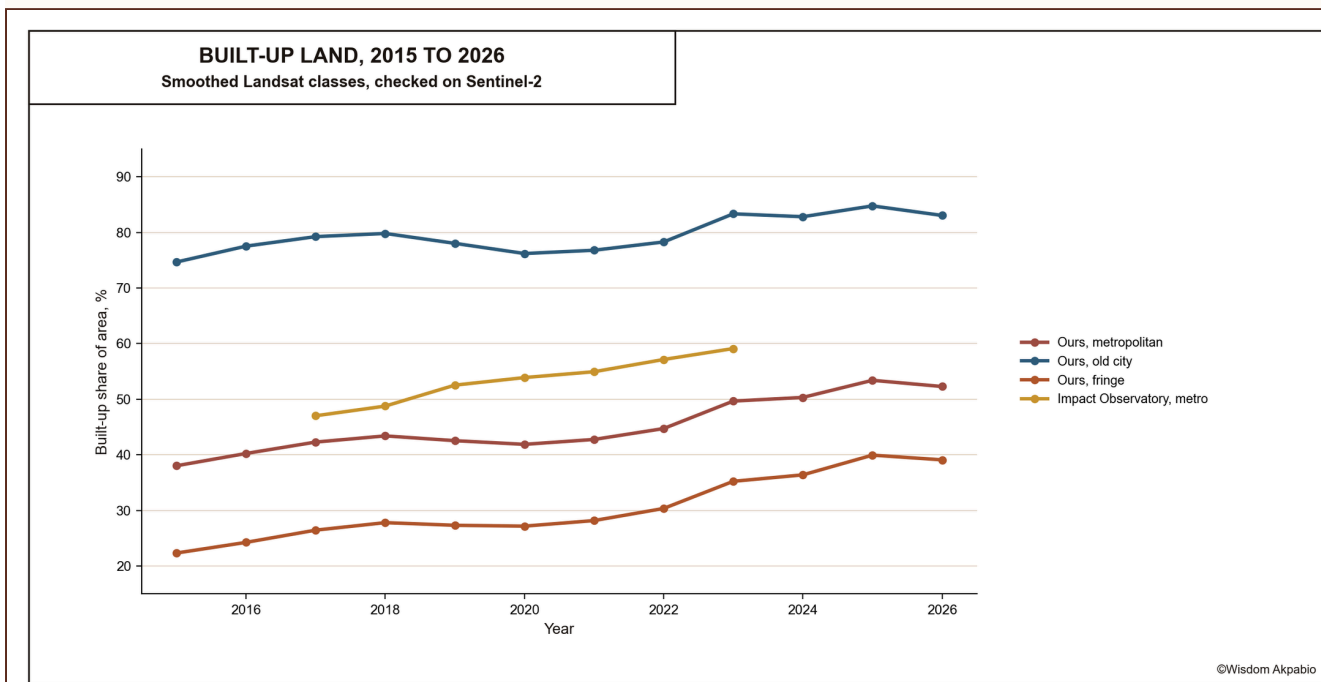


Figure 8. Built-up share of area, 2015 to 2026. The old city is already mostly built. The metropolitan rise is the fringe filling in. Impact Observatory sits higher than our Landsat maps and moves in the same direction.

For each hexagon the comparison is 2015–17 against 2024–26. Heat is the hexagon’s hot-season

surface temperature minus that year’s rural-ring median, so a hot or cool year does not pass for change. The fair comparison is farmland that was built over against farmland that stayed open.

Table 7. Change in daytime surface heat, farmland that was built over minus farmland that stayed open. Built over: under 25% built-up in the early period, at least 50% in the late period. Stayed open: under 25% in both.

Land cover used	Period	Built-over hexagons	Built-up then → now	Built over minus stayed open
Impact Observatory (Sentinel-2, independent)	2017-18 → 2022-23	417	6% → 83%	−0.3 °C (median −0.6)
Ours (Landsat)	2017-18 → 2022-23	117	15% → 67%	−1.5 °C (median −1.6)
Ours (Landsat)	2015-17 → 2024-26	378	13% → 72%	−1.4 °C (median −1.5)

Farmland that became city cooled by day relative to farmland that stayed open, whichever land cover is used. The size depends on it. The Landsat classes and the heat come from the same scenes, and on the same years those maps flag far fewer hexagons than Impact Observatory, the ones whose Landsat signal changed most; that inflates the effect. The independent figure, **0.3 to 0.6 °C**, is the one to quote, with ours as an upper bound. Plate 4 shows where: Gayawa, Rangaza, Kadawa and Tudun Fulani in Ungogo; Garun Gawa, Kureken Sani, Danbare and Mariri in Kumbotso. The 378 built-over neighbourhoods on our maps are home to **320,000** people today.

Read with section 5.5, this is a daytime result. The same walls that cool the fringe at 10:30 are the ones that keep the old city warm at 01:30.

6. Discussion

Kano behaves like other dry cities: by day in the dry season its built fabric is cooler than the bare fields around it, and by night it is warmer. Earth and cement compounds, narrow shaded lanes, courtyard trees and irrigated gardens along the rivers keep the daytime surface below that of open, sunlit, dry soil. The same walls and roofs store that heat and release it after dark. The anchor’s advice, more vegetation per plot, is right for Bangalore and weak for Kano in March. What cools here by day is shade and moisture, not greenness as a satellite measures it.

Two findings matter for planning, and they point at different places. By day it is the edge. Kano’s growth in Ungogo and Kumbotso is building onto the hottest daytime ground in the metropolitan area, and the people already there are the ones farthest from clinics and schools. The walking paper found the later population living in Ungogo and Kumbotso while services stayed in the old city. This paper adds that the same families are on the hottest ground at midday. Gayawa was already the clearest miss on walking time. Karo, Yada Kunya, Rangaza and Fanisau are where that miss sits on hot ground.

By night it is the old city. The dense wards that shelter people at 10:30 hold the most heat at 01:30, in the months when nights matter most for rest. Neither is visible from a single morning scene, which is what the anchor relied on.

The yearly maps (Figure 7) make the growth less abstract. Built-up cover in the old city was already high in 2015. The metropolitan rise from 38% to 52% is Ungogo and Kumbotso filling in, which is also where the independent change test finds daytime cooling. That cooling is not a reason to call the new fringe comfortable. It is a reason not to treat every new roof as a heat island.

7. Limitations

These are surface temperatures, not air temperatures and not heat stress. There is no humidity or wind here. The night-time signal is at 1 km from MODIS, not at the 30 m of the daytime maps, so a night-time ranking of wards is coarser than a daytime one.

The composition analysis uses WorldCover 2021. WorldCover's vegetation and cropland classes are not a field survey, and the anchor's composition table is ambiguous; the gradient is the robust version. The yearly land cover is our own classification, 85.6% on pure, stable cells and lower on mixed edge pixels. Grass and shrub is the class that slips (F1 0.71), mostly into cropland, which is the confusion a dry-season Sahel map is expected to make. The yearly maps share scenes with the heat, so the change effect is quoted from the independent Impact Observatory maps.

Kwachiri's hot pixels include the airport runway. That is real hot ground. It is not a neighbourhood in the sense of Table 5. Fanisau, Rangaza and Karo are.

Scene counts vary from one to six a year, so year-to-year absolute temperatures reflect which days were clear. Only contrasts within a date are compared across years. The three scenes that break the cool-island pattern are all in May, after rain. A paper that had used only April, as the anchor did, would have missed them and also missed the night.

Population and age-sex rasters are modelled, not counted, and the age structure is uniform across the city. The walking times are the walking paper's: 5 km/h on a mapped carriageway. OpenStreetMap has no sidewalk tag on these edges. Distance on that line is not the same as a street someone will walk in the Kano heat, and it is not how the longer trips in Ungogo actually happen.

8. Policy recommendations

Two places, two times of day. Mixing them into one "cool the city" programme will spend money where the surface is already cool at 10:30 and leave the people on the hot edge as they are.

By day the work is in Ungogo and Kumbotso. Karo, Yada Kunya, Rangaza, Fanisau and Chalawa are the wards that are both hot and far. Shade and moisture on those streets, and clinics and schools on streets that go through, are the same backlog the walking paper already named. A new clinic in Dala will not shorten those walks and will not sit on that hot ground. Gayawa is still the clearest miss on access; the growth in Gayawa, Rangaza, Kadawa and Tudun Fulani is also where farmland became city.

By night the work is in the old city. The dense wards that show blue on Plate 1 are the ones that hold heat after dark. Courtyards, through-ventilation and whatever moisture those compounds already keep are the relevant fabric. Treating Dala or Gwale as a daytime heat island, because that is what a Bangalore reading expects, will plant the wrong intervention in the wrong season.

The 30% vegetation rule does not travel to Kano in March. What the satellite calls green then is often dry cropland, and that cropland is the hot surface. Irrigation, shade and open water track the cool ground. A tree-cover target copied from a humid city will look busy on a land-use plan and do little to the March surface.

Building on farmland cools the daytime surface relative to farmland that stays open. That is not an argument for unplanned growth. The same walls warm the night, and the people moving onto that edge are the ones already far from services. Quote the independent 0.3 to 0.6 °C if the change is discussed at all, and say that it is a morning result.

None of this is air temperature. A ward chairman who wants to know whether people can sleep still needs humidity, wind and a night-time map finer than 1 km.

9. Conclusions

Metropolitan Kano is a daytime cool island in the hot season and a heat island at night. The cool ground at 10:30 is the built city. The hot ground is the farmland edge, and almost everyone who lives on that edge is also more than fifteen minutes' walk from clinics and schools. After dark the old city is the warm ground. Built-up land grew from 38% to 52% of the outline between 2015 and 2026; farmland that became city cooled by day, on independent land cover, by 0.3 to 0.6 °C. Greenness does not predict the March surface. The Bangalore morning scene does not describe this city.

Plate 1. Hot-season surface temperature on 200 m hexagons, pooled 2015 to 2026. Blue is cooler ground, red is hotter.

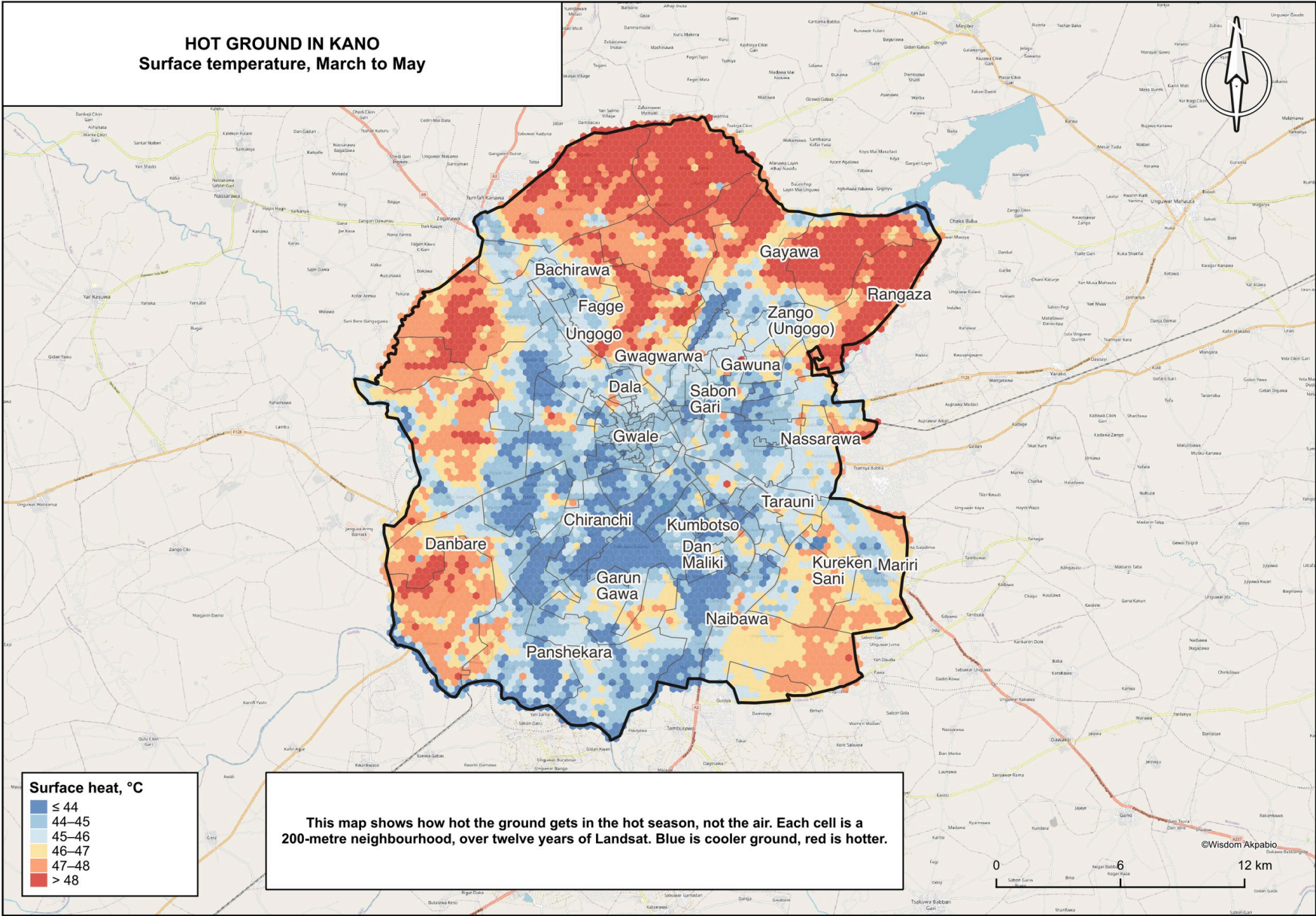


Plate 2. Land cover in Kano. Most common ESA WorldCover 2021 class in each 30 m cell. Built-up 45% of the area, cropland 40%. The yearly series in Figure 7 is our own classification; this plate is the 2021 reference.

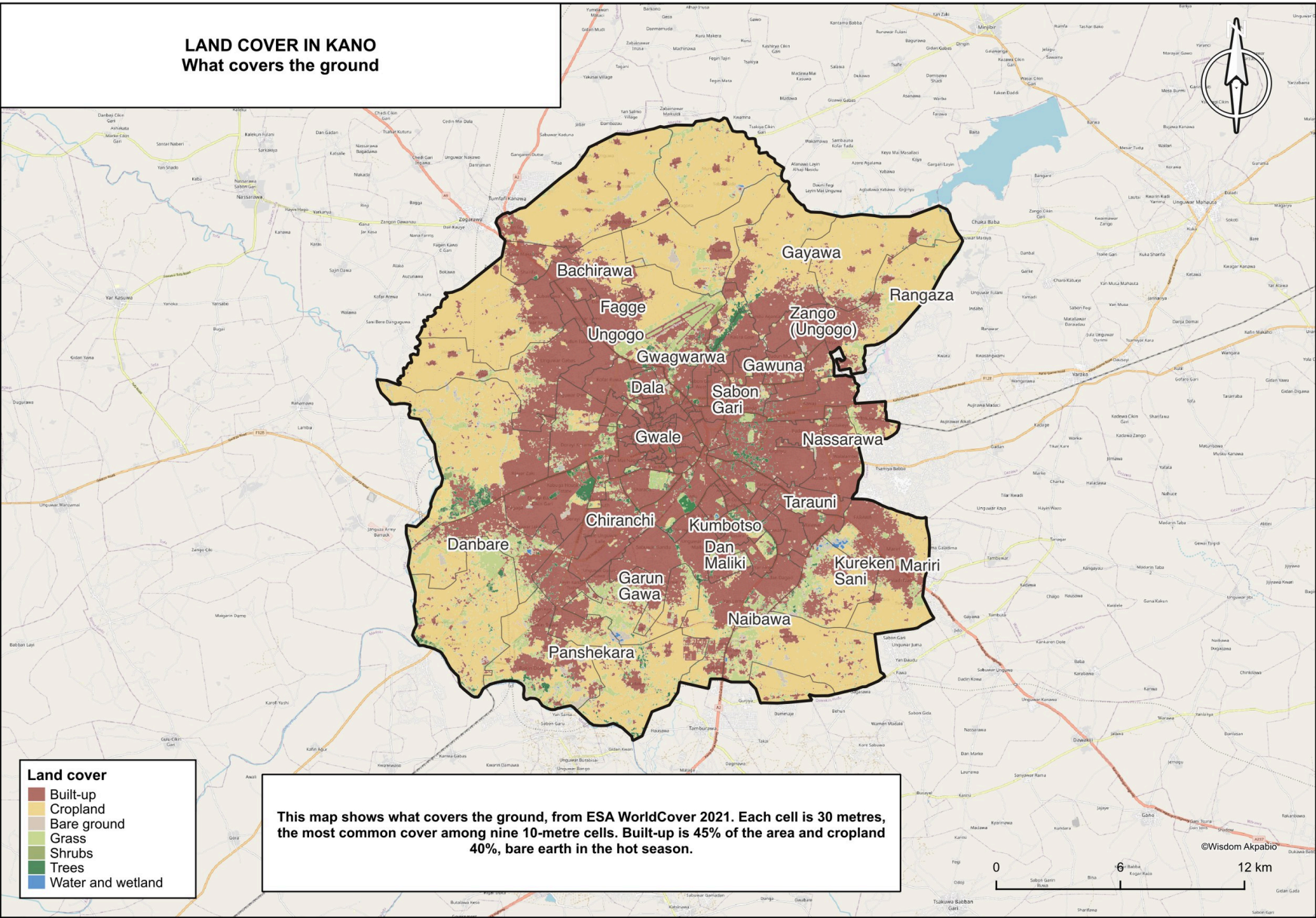


Plate 3. Hot ground and long walks. Hot is the hottest fifth of populated neighbourhoods; far is more than a 15-minute walk to five clinics and five schools. Dark red is both: 261,000 people.

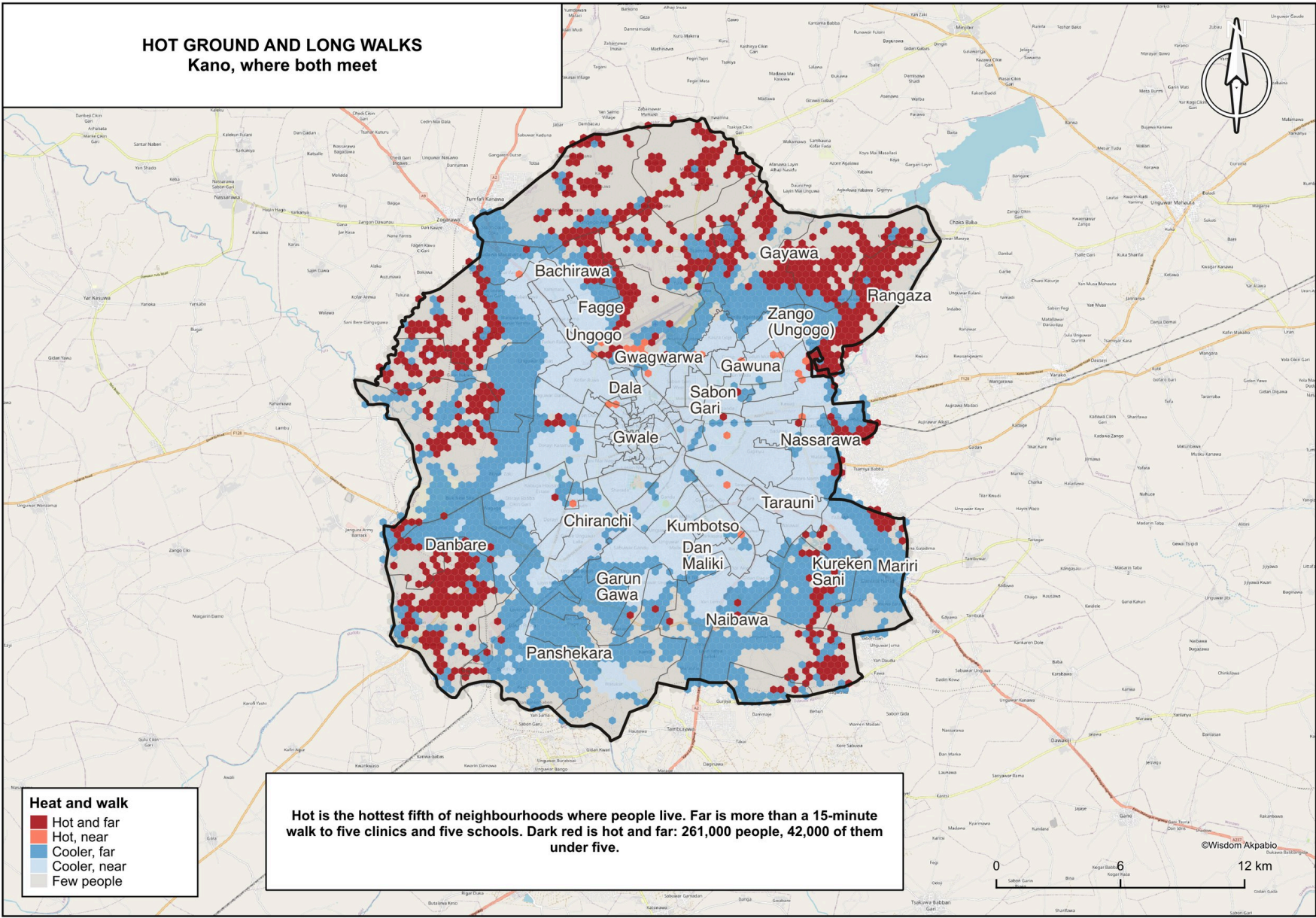
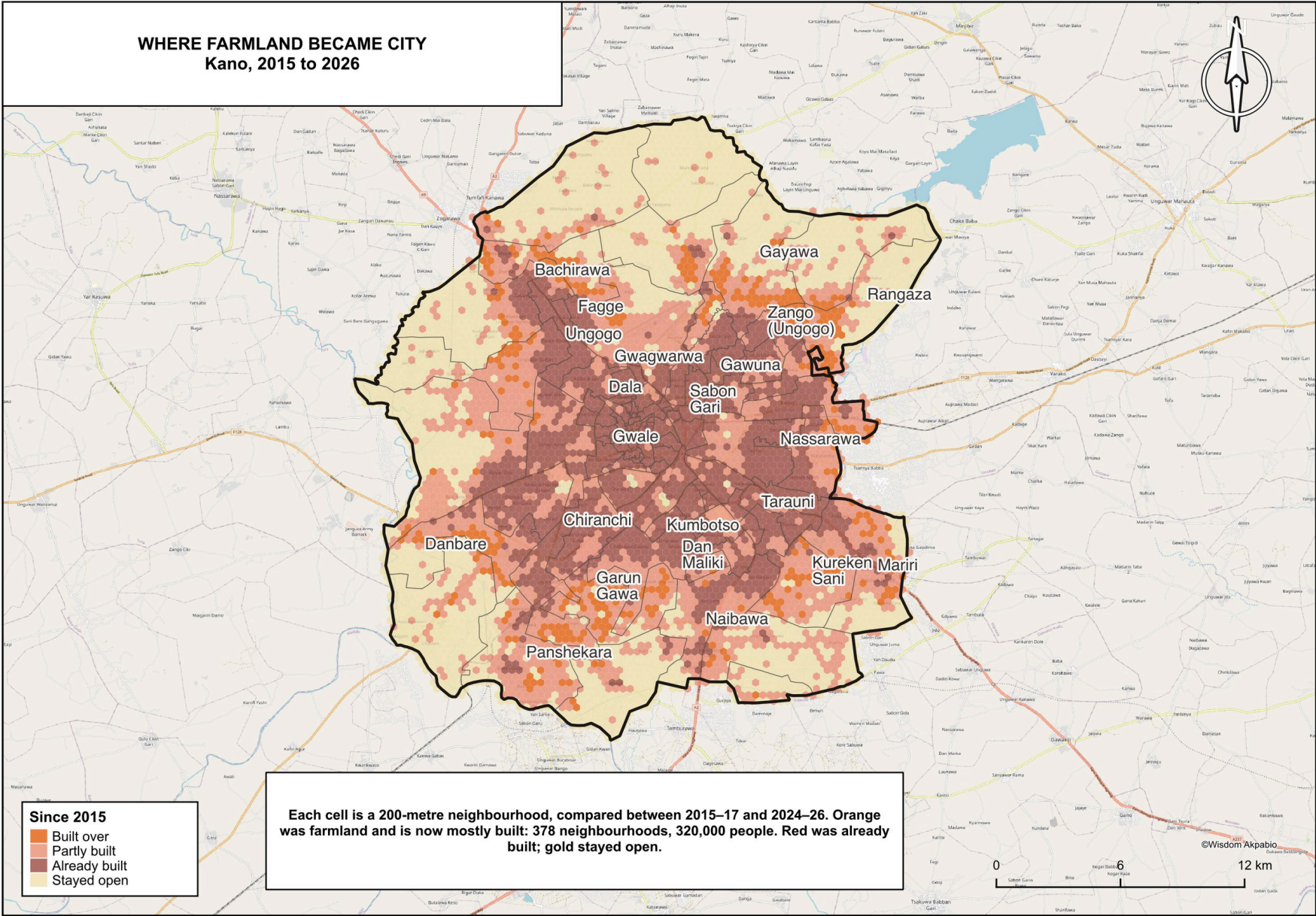


Plate 4. Where farmland became city. Neighbourhoods compared between 2015-17 and 2024-26 on our Landsat land cover. Orange was farmland and is now mostly built.



Data availability

Write-up and brief: `notes/results.md`, `notes/brief.md`, `docs/paper/kano_surface_heat.pdf`.
Quoted tables: `data/processed/lst_by_scene.csv`, `lst_series.csv`, `lst_by_lga.csv`,
`composition_gradient.csv`, `lulc_accuracy.csv`, `lulc_confusion.csv`, `lulc_shares_by_year.csv`,
`modis_summary.csv`. Maps: `maps/kano_study_area.png`, `maps/kano_heat_plate.png`,
`maps/kano_lulc_plate.png`, `maps/kano_heat_walk_plate.png`, `maps/kano_change_plate.png`. Charts:
`charts/lulc_years.png`, `charts/lst_years.png`, `charts/lulc_accuracy.png`,
`charts/built_up_share.png`, `charts/cool_island_by_year.png`, `charts/built_cells_cool.png`,
`charts/cool_day_warm_night.png`. Code: <https://github.com/whizialfa/kano-surface-heat>. Landsat,
MODIS, ESA WorldCover, Impact Observatory, GRID3 and OpenStreetMap keep their original licences.
Quote the cool island from March–April scene counts, and the change effect from Impact Observatory.

References

- Akpabio, W. (2026). *Fifteen minutes on foot: walking to clinics and schools in Lagos, Kano, Ibadan, Abuja and Port Harcourt*.
<https://whizialfa.github.io/15-min-cities-nigeria/>
- GRID3 (2025). Nigeria gridded population estimates v3.0 and age-sex structures.
- Karra, K. et al. (2021). Global land use / land cover with Sentinel-2 and deep learning. IGARSS 2021 (Impact Observatory annual LULC, v02).
- Lazzarini, M., Marpu, P.R. & Ghedira, H. (2013). Temperature-land cover interactions: the inversion of urban heat island phenomenon in desert city areas. *Remote Sensing of Environment* 130, 136–152.
- Oke, T.R. (1982). The energetic basis of the urban heat island. *Quarterly Journal of the Royal Meteorological Society* 108, 1–24.
- Peng, S., Piao, S., Ciaia, P., Friedlingstein, P., Ottle, C., Bréon, F.-M., Nan, H., Zhou, L. & Myneni, R.B. (2012). Surface urban heat island across 419 global big cities. *Environmental Science & Technology* 46, 696–703.
- Ramachandra, T. V., Rana, R. S., Vinay, S. & Aithal, B. H. (2025). Urban heat island linkages with the landscape morphology. *Scientific Reports* 15, 24485. <https://doi.org/10.1038/s41598-025-09141-5>
- Rasul, A., Balzter, H. & Smith, C. (2015). Spatial variation of the daytime Surface Urban Cool Island during the dry season in Erbil, Iraqi Kurdistan, from Landsat 8. *Urban Climate* 14, 176–186.
- U.S. Geological Survey. Landsat 8–9 Collection 2 Level-2 Science Products.
- Voogt, J.A. & Oke, T.R. (2003). Thermal remote sensing of urban climates. *Remote Sensing of Environment* 86, 370–384.
- Wan, Z., Hook, S. & Hulley, G. (2021). MODIS/Terra and MODIS/Aqua Land Surface Temperature/Emissivity 8-Day L3 Global 1 km SIN Grid V061 (MOD11A2, MYD11A2). NASA EOSDIS Land Processes DAAC.
- Zanaga, D. et al. (2022). ESA WorldCover 10 m 2021 v200. <https://doi.org/10.5281/zenodo.7254221>

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